

SCHEMA

docs/coverage/SCHEMA.md — COVERAGE_LEDGER.md row format

Round 68 creation. This document defines the schema for COVERAGE_LEDGER.md. Schema changes require an entry in the audit-trail table at the bottom of COVERAGE_LEDGER.md AND a bump of SCHEMA_VERSION below.

SCHEMA_VERSION = 1.0.0

1. Row format

Every data row in COVERAGE_LEDGER.md has exactly 11 columns:

| Submodule | Feature | Platform | I1 anti-bluff | I2 e2e-working | I3 doc

1.1 Column definitions

#	Column	Type	Required	Description
1	Submodule	string	yes	Path-from-repo-root identifier per docs/improvements/submodule_owned.txt (e.g. submodules/auto_temp, helix_qa, challenges).
2	Feature	string	yes	Short description of the capability under audit (e.g. grid-search temperature tuning, OpenCode CLI adapter). May be whole-module for submodules audited as one unit.
3	Platform	string	yes	Comma-separated platform list from {linux, macos, windows, ios, android, aurora-os, harmony-os, containers, headless}. all-platforms permitted only for platform-agnostic Go code.
4-9	I1..I6	status	yes	

#	Column	Type	Required	Description
				CONST-048 invariant status per §2 vocabulary below.
10	Overall	status	yes	Rollup: PASS only if every I1..I6 is PASS or N/A; otherwise PARTIAL or UNCONFIRMED: . Generator computes mechanically.
11	Notes	string	yes	MUST include either (a) CONTINUATION round reference (e.g. round 37 wired Real backends) for any PASS marker, OR (b) free-text gap description for non-PASS cells. Blank Notes column on a PASS row is a CONST-035 PASS-bluff.

1.2 Row uniqueness

The tuple (Submodule, Feature, Platform) MUST be unique. Duplicate rows are a violation; collapse them by merging Notes.

1.3 Submodule coverage requirement

Every entry in `docs/improvements/submodule_owned.txt` MUST appear in at least one row. The generator's `--check` mode (`scripts/generate-coverage-ledger.sh --check`) exits non-zero if any owned submodule has zero ledger rows — this is the round 68 enforcement gate for CONST-048's "rows that quietly omit a platform are CONST-048 violations" mandate.

2. Status vocabulary

Closed set; values outside this list are violations.

Symbol	Semantics	Evidence requirement
PASS	Invariant met in current cycle with captured runtime evidence.	Notes column MUST cite CONTINUATION round OR pasted evidence path.
PENDING_FORENSICS:	Work landed in source/tests but evidence not yet captured into ledger.	Notes SHOULD cite the commit SHA that introduced the work.
OPERATOR-BLOCKED:	Cannot complete without operator action (real LLM	Notes MUST cite the operator dependency per §11.4.21 audit pattern.

Symbol	Semantics	Evidence requirement
	key, hardware, real network).	
UNCONFIRMED:	Not yet audited. Default for new rows.	Notes SHOULD describe what audit is needed.
SKIP-OK: #<ticket>	Intentionally not covered.	Notes MUST cite the ticket explaining the skip.
PARTIAL	Some platforms / scenarios covered but not all.	Notes MUST enumerate which platforms/ scenarios pass vs. fail.
N/A	Invariant does not apply (e.g. I2 e2e-working for a build-tooling-only submodule).	Notes SHOULD justify the N/A.

Default cell status is UNCONFIRMED: — never PASS. PASS without (a) CONTINUATION reference or (b) pasted evidence is a CONST-035 PASS-bluff equivalent in severity to a false-success test result (per Article XI §11.9).

3. Cross-reference points

The ledger does NOT duplicate evidence — it points to canonical sources.

Invariant	Canonical source	Cross-check command
I1 anti-bluff	CONTINUATION.md round narrative + commit SHA	<code>grep -n "round <N>" docs/CONTINUATION.md</code>
I2 e2e-working	Real-infra test logs (/tmp/coverage-*) + CONTINUATION captured evidence	<code>grep -rn "real <provider>\ wire evidence" docs/CONTINUATION.md</code>
I3 doc-match	User manual + docs/COMPLETE_*.md references	<code>grep -rn "<feature>" docs/COMPLETE_*</code>
I4 no-issues	Issues.md / Issues_Summary.md (if maintained) + source-tree TODO(round-N) / BUG #NN: markers	<code>grep -rn "TODO(round-\ BUG #" <submodule>/</code>
I5 doc-sync	Last-modified timestamp of user manual + submodule docs/tree	<code>find <submodule>/docs -newer <submodule>/cmd -type f</code>
I6 4-layer-tests	Pre-build gate (make verify-compile) + post-build (make test) + runtime (challenges/e2e) + paired mutation (search for	<code>find <submodule> -name "*_test.go" \ xargs grep -l "SKIP-OK\ TestMutation"</code>

Invariant	Canonical source	Cross-check command
	<code>_test.go + sibling mutation gate)</code>	

3.1 No-bugs cross-check (I4) — HelixCode-specific note

HelixCode does not currently maintain a root-level `Issues.md` tracker (per round 32 audit). I4 cross-check therefore relies on:

1. Source-tree markers: `grep -rn "TODO(round-\|BUG #\|FIXME(round-\|XXX(round-" <submodule-path>/` — non-empty result = I4 cannot be PASS.
2. `docs/CONTINUATION.md` Known issues / Deferred work sections — if the submodule is named in a known-issue paragraph, I4 cannot be PASS.
3. Open BLUFFS in `docs/issues/BLUFFS.md` (if file exists) naming the submodule — I4 cannot be PASS.

Future rounds may introduce a per-submodule `Issues.md` tracker per CONST-057 / §11.4.16; the generator script will adopt it automatically when present.

4. Status rollup (column 10 “Overall”)

Mechanical rollup computed by `scripts/generate-coverage-ledger.sh --refresh`:

```
if all(I1..I6 in {PASS, N/A}):           Overall = PASS
elif any(I1..I6 in {OPERATOR-BLOCKED:}): Overall = OPERATOR-BLOCKED:
elif any(I1..I6 in {PENDING_FORENSICS:}): Overall = PENDING_FORENSICS:
elif any(I1..I6 in {PARTIAL}):           Overall = PARTIAL
elif any(I1..I6 in {SKIP-OK: ...}):       Overall = PARTIAL (any skip is a
else:                                     Overall = UNCONFIRMED:
```

The generator NEVER promotes individual invariant cells — it only computes the Overall rollup from whatever cells the operator has hand-marked.

5. Anti-bluff guarantees (cascade from CONST-035 / Article XI §11.9)

1. **No PASS-by-default.** Every cell starts UNCONFIRMED: until hand-marked with evidence reference.
2. **No silent omissions.** Every owned submodule has at least one row; `generate-coverage-ledger.sh --check` fails otherwise.
3. **Evidence-required Notes.** PASS in any cell requires Notes content; blank Notes on a PASS row = violation.
4. **Status vocabulary is closed.** Values outside the §2 list = violation.
5. **Schema version bump.** Any change to column count / column order / new status symbol requires `SCHEMA_VERSION` bump + audit-trail entry.
6. **Cross-ledger consistency.** Feature names used here MUST be discoverable in either `docs/coverage/ledger.md` (F01..FNN catalogue) OR `docs/user_manual/ZERO_BLUFF_USER_MANUAL.md`; otherwise the feature is a phantom and the row is a violation.

6. Regeneration / refresh contract

`scripts/generate-coverage-ledger.sh` has two modes:

- `--scaffold` (default): emit a baseline ledger with one row per owned submodule, all invariants `UNCONFIRMED:`. Used when bootstrapping a fresh ledger.
- `--check`: walk the current `COVERAGE_LEDGER.md`, assert each owned submodule has ≥ 1 row, assert schema invariants, assert every PASS has Notes. Exit non-zero on any violation.

Neither mode auto-promotes cells. Neither mode rewrites operator-marked PASS values.

7. Audit-trail discipline

Schema changes append a row to `COVERAGE_LEDGER.md`'s audit-trail table with: - Date (ISO YYYY-MM-DD) - Author (agent ID or operator name) - Round (close-outNN or round-NN) - Schema version bump (e.g. `1.0.0` $\rightarrow$ `1.1.0`) - One-line rationale

Initial schema (round 68) entry: see `COVERAGE_LEDGER.md` audit trail.

Sources verified 2026-05-29: internal-governance doc — no third-party-service operator instructions to cross-reference. This document defines the row/column schema and status vocabulary for HelixCode's own `COVERAGE_LEDGER.md`; it references only internal constitutional anchors (CONST-035/048, §11.4.x) and HelixCode's own generator scripts. No external service, library, or version pin appears, so there is nothing to WebFetch against vendor docs. Version authority per CLAUDE.md §3.1 (confirmed in-tree): inner module `go 1.26`, root `go 1.25.2`. Reviewed against the live tree on this date; no corrections required.